

Question 1

A negative-feedback amplifier has a closed-loop gain $A_f = 100$ and open-loop gain $A = 10^5$.

- What is the feedback factor β ?
- If a manufacturing error results in a reduction of A to 10^3 , what closed-loop gain results?
- What is the percentage change in A_f corresponding to this factor of 100 reduction in A ?

Question 2

In a particular amplifier design, the β network consists of a linear potentiometer for which β is 0.0 and one end, 1.0 at the other end, and 0.5 in the middle. As the potentiometer is adjusted, find the three values of closed-loop gain that result when the amplifier open-loop gain that is (a) 1 V/V, (b) 10 V/V, (c) 100 V/V, (d) 10,000 V/V.

Question 3

A newly constructed feedback amplifier undergoes a performance test with the following results: With the feedback connection removed, a source signal of 2 mV is required to provide a 10-V output; with the feedback connected, a 10-V output requires a 200-mV source signal. For this amplifier, identify values of A , β , $A\beta$, the closed-loop gain, and the amount of feedback.

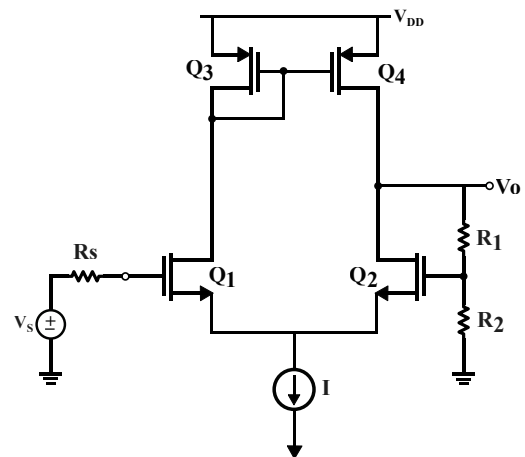
Question 4

For the negative-feedback loop, find the loop gain $A\beta$ for which the sensitivity of closed-loop gain to open-loop gain [i.e., $(dA_f/A_f) / (dA/A)$] is -20 dB. For what value of $A\beta$ does the sensitivity become 1/2.

Question 5

The current-mirror-loaded differential amplifier in Fig., has a feedback network consisting of the voltage divider (R_1, R_2), with $R_1 + R_2 = 1 \text{ M}\Omega$. The devices are sized to operate at $|V_{OV}| = 0.2 \text{ V}$. For all devices, $|V_A| = 10 \text{ V}$. The input signal source has a zero dc component. Assume $I = 200 \mu\text{A}$.

- Find the loop gain $A\beta$ and hence the value of A .
- Find the values of R_1 and R_2 that result in a closed-loop gain of exactly 5 V/V.



Question 6

In the given circuit, all three MOSFETs are biased to operate at $g_m = 4 \text{ mA/V}$. The resistance values are $R_{D1} = 10 \text{ k}\Omega$, $R_{D2} = 10 \text{ k}\Omega$, $R_{S1} = 100 \text{ }\Omega$, $R_{S2} = 100 \text{ }\Omega$. Calculate the value of R_F that results in a closed-loop gain is 10 V/V .

